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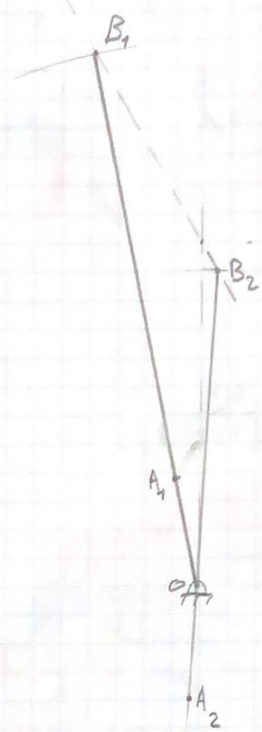
Assignment "1"
Mechanics

1) a) Scale 1:2

b) Stroke $= 4.4 \times 2 = 8.8$ cm



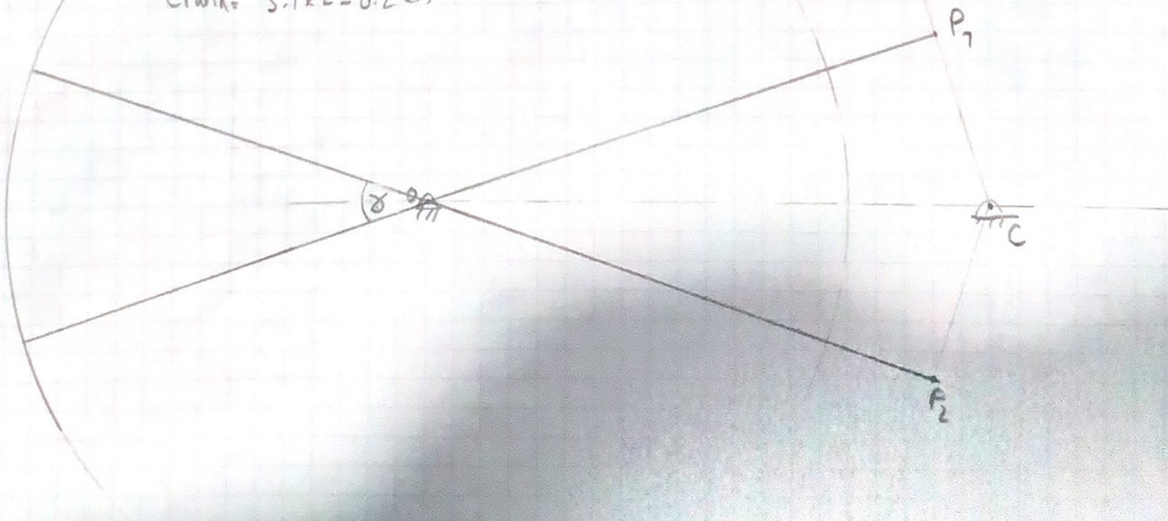
2) Scale 1:2

$\frac{\theta_c}{\theta_r} = 1.5$, $\theta_c + \theta_r = 360°$

$\theta_c = 216°$, $\theta_r = 144°$

$\alpha = 37°$, Stroke $= 7.5 \times 2 \times \frac{37\pi}{180°} = 9.686$ cm

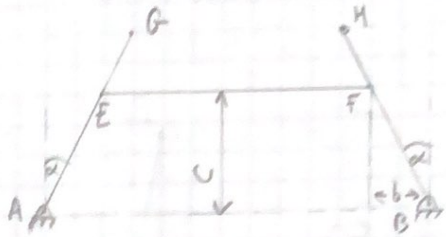
Crank $= 3.1 \times 2 = 6.2$ cm



3) $\tan(\alpha+\phi) = \frac{b+x}{c}$

$\tan(\alpha-\phi) = \frac{b-x}{c}$

$\tan(\alpha) = \frac{b}{c}$



$$\tan(\alpha+\phi) = \frac{\tan\alpha + \tan\phi}{1-\tan\alpha\tan\phi}$$

$$\frac{b+x}{c} = \frac{\frac{b}{c} + \tan\phi}{1-\frac{b}{c}\tan\phi} = \frac{b+c\tan\phi}{c-b\tan\phi}$$

$$\tan\phi = \frac{cx}{b^2+c^2+bx} \quad , \therefore \tan\theta = \frac{cx}{b^2+c^2-bx}$$

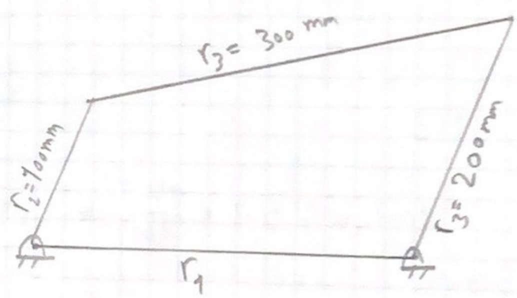
$$\therefore \cot\phi - \cot\theta = \frac{a}{d}$$

$$\therefore \frac{b^2+c^2+bx}{cx} - \frac{b^2+c^2-bx}{cx} = \frac{a}{d} \quad , \quad \frac{2bx}{cx} = \frac{a}{d} \quad , \quad \frac{2b}{c} = \frac{a}{d}$$

4) a) $S+L \leq P+Q$

1: r_1 is L $\rightarrow 100 + r_1 \leq 300+200$

$r_1 \leq 400$



2: r_1 is P $\rightarrow 100+300 \leq r_1 + 200$

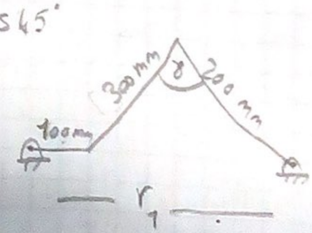
$200 \leq r_1$

$\therefore$ to get crank rocke r_1 must be between 200 & 400

$$200 \leq r_1 \leq 400$$

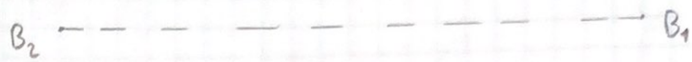
b) $(r_1 - 100)^2 = 300^2 + 200^2 - 2(300)(200)\cos 45^\circ$

$r_1 = 312.478$ mm



5) $\theta_c = 1.25\,\theta_r$, $\theta_c + \theta_r = 360^\circ$

$\theta_c = 200^\circ$, $\theta_r = 160^\circ$, $\delta = 20^\circ$



6) $\theta_c = 1.25\,\theta_r$, $\theta_c + \theta_r = 360^\circ$

$\theta_c = 200^\circ$, $\theta_r = 160^\circ$

$\delta = 20^\circ$